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RESEARCH ARTICLE

SLGAN-EID: Generative Adversarial Network-Based Skin Lesion Image Enhancement With Improved Diagnosis

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ABSTRACT Skin cancer is one of the most common malignancies worldwide and its timely detection is crucial. Its early detection substantially improves the survival rate of the patient. Accurate diagnosis depends on high-quality dermoscopic images. However, image quality and diagnostic accuracy are degraded by different factors such as noise artifacts, hair occlusion, air bubbles, and uneven lighting. In addition, significant inter-observer variability among clinicians further limits diagnostic consistency. Although existing image restoration techniques improve perceptual quality but unconsciously compromise important diagnostic features required for accurate lesion identification, thus providing an inherent trade-off between visual enhancement and clinical utility. This paper introduces the Skin Lesion Generative Adversarial Network for Enhancement with Improved Diagnosis (SLGAN-EID) which introduces key methodological contributions. Firstly, we present a diagnostic-aware dual-discriminative network, quality discriminator (D_1) which is used to ensure perceptual fidelity, and another is a diagnostic discriminative network (D_2), used to explicitly preserve clinically meaningful features, i.e., Asymmetry, Border irregularity, Color variation, Diameter (ABCD). Second, we add a dermoscopy-specific three-scale attention mechanism (different from conventional attention mechanisms) that separates dermoscopic artifacts and features (reticular lines, globules, and dots) of clinical significance that allow artifact reduction without degradation of the characteristics. We also include a new preservation of clinical feature loss ($\mathcal{L}_{diagnostic}$), which directly optimizes for dermatologist-relevant features instead of only pixel-based reconstruction metrics. The generator uses a U-Net backbone similar with multi-scale attention, whereas VGG-19 captures similarities between hierarchical representations across perceptual similarity. The proposed SLGAN-EID framework is tested on ISIC 2019 (25,331 images, 8 classes) and HAM10000 (10,015 images, 7 classes) datasets, achieved 94.3% classification accuracy with 0.923 F1-score, improving PSNR from 22.14 dB to 32.87 dB and SSIM from 0.621 to 0.863. Furthermore, a clinical study involving expert dermatologists reports a 27% increase in diagnostic confidence. The proposed framework also supports real-time deployment, achieving an average processing time of 112 ms per image on a standard workstation, thereby accelerating diagnostic workflows without sacrificing accuracy.

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INDEX TERMS Generative adversarial networks, skin lesion enhancement, dermoscopy, medical image processing, deep learning, diagnostic imaging, attention mechanisms.

I. INTRODUCTION

Skin cancer is an important health challenge in the world, and according to the World Health Organization (WHO), the disease accounts for more than 1.5 million new cases annually and melanoma alone claims approximately 60,000 lives per year around the world [1]. The five-year prognosis of melanoma at the initial stages is over 99%, but when the disease is found at the later stages of its progression, this figure decreases to 32% [2]. Dermoscopy has become the dominant non-invasive form of diagnoses, allowing the visualization of the underlying skin structure not visible by the naked eye and has increased diagnostic sensitivity by 10%-27% compared to clinical examination alone [3]. Nevertheless, the image quality limitations inherent in the clinical utility of dermoscopy are fundamentally limiting the diagnostic accuracy and treatment decisions.

A. CHALLENGES IN DERMOSCOPY IMAGE ANALYSIS

The modern dermoscopy devices often generate images, which are deteriorated by various artifacts making the lesions identification difficult. The most common type of artifact is hair occlusions, which covering 40%-60% of dermoscopic images, disrupt morphological features including pigment network patterns and lesion margins, which are critical in diagnosis [5]. Local intensity distortions caused by air bubbles and gel artifacts may resemble or conceal true dermoscopic structures, whereas high-contrast interference patterns generated by marker and ruler artifacts of measurement protocols in clinical use impede absolute changes in absolute intensity [4]. Light source irregularities due to unequal placement produce color distributions variability and reflective light as well as produce intensity variations and light reflections dissimilar which are key diagnostic features of the Asymmetry, Border irregularity, Color variation, Diameter (ABCD) assessment model [6]. These artifacts significantly degrade dermoscopic image quality and can lead to misdiagnosis. Studies have shown that their presence increases diagnostic error rates by approximately 15–25%, while moderate artifact contamination can result in up to a 12% rise in false-negative melanoma detections. In cases of severe artifact contamination, false-negative rates have been reported to reach as high as 18%, highlighting the critical impact of artifacts on diagnostic accuracy [9]. Babu and Chandramohan in [7] incorporated GAN-based data augmentation, and a Convolutional Neural Network (CNN) classifier to boost feature learning and enhance generalization with a variety of dermoscopic images. The joint optimization of performance between image generation and classification is presented as a hybrid loss, and an attention mechanism is used to further optimize the discriminative feature extraction. The possibility of GAN-enhanced deep learning systems to enhance reliability and scalability of automated skin cancer diagnosis. Such statistics point to the urgent necessity to develop well-representative image enhancement strategies that can also reevaluate original diagnostic use without imposing new artificial components that would lead to poor clinical decision-making.

B. LIMITATIONS OF CONVENTIONAL ENHANCEMENT APPROACHES

Conventional methods, such as histogram equalization, Contrast Limited Adaptive Histogram Equalization (CLAHE) and morphological filtering, are based on rules of thumb algorithm construction and statistical assumptions that poorly adapt to the complex and non-uniform nature of the dermoscopy artifacts [8]. Although these methods can progressively obtain moderate advances in measurable of the quality of perception, they work without understanding clinically significant aspects which make them useless for practical usage [10]. Both wavelet-based denoising and anisotropic diffusion filtering are more advanced, but still fail to identify pathological characteristics and artifacts of acquisition, and also less optimal for dermoscopic image enhancement, potentially distorting diagnostic evidence [11].

Morphological-based and inpainting hair removal algorithms are specialized to only one type of artifact but fail to cross-contaminated hair-skin edges and do not unify the parameters of dermoscopy image degradation in a multi-factorial way [12]. Moreover, the traditional methods treat enhancement and diagnosis as a strictly sequential and independent task that does not allow optimizing the parameters of enhancement towards downstream diagnostic purposes and establishes an inherent discontinuity between image processing and clinical utility [13].

C. GENERATIVE ADVERSARIAL NETWORKS (GANS) FOR MEDICAL IMAGE ENHANCEMENT

GANs have transformed the medical image processing field and widely used in hundreds of different applications in image generation, synthesis, and analysis settings [13]. Adversarial training is an application of Deep Learning (DL) paradigm that facilitates the learning of complex data distributions without explicitly modeling the data, thus GANs are especially adapted to medical imaging, where ground truth labels are usually scarce [14].

Other recent advances in GAN applications encompass super-resolution images reconstruction [15], removal of artifacts, and production of synthetic data to promote training enhancement immunity [16], as well as recognition of lesion [17]. Nevertheless, the majority of the existing solutions are aimed at the general medical imaging activities but not at the dermoscopy specific applications, which restricts their usefulness in the skin lesion analysis.

The discriminator element of GAN systems is a learned perceptual quality inspector that is more suitable with human visual response, relative to the conventional pixel-based metrics such as mean squared error. This perception learning ability facilitates the creation of superior images that are natural and clinically realistic instead of having the over-sharpening, halo effects, or unnatural color changes of traditional enhancement [18], [37], [38]. The latest developments in conditional GAN designs and attention systems have also expanded these functionality with spatially adaptive processing functionality that allows the selective targeting of damaged areas without destroying diagnostic content in healthy areas [25].

Existing GAN-based dermoscopic image enhancement methods exhibit signal limitation restricting their use in practical clinical applications. These approaches are mainly optimized based on perceptual quality metrics including Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Metric (SSIM) which have a very low correlation with diagnostic utility and clinical decision-making accuracy [19]. Lack of diagnostic awareness loss function and conditions may provide high scores on perceptual quality with less morphological details that are vital to the accurate classification of lesions. It is therefore, evident that improvement in visual quality may lose the vital diagnostic information [20].

The latest dermoscopy image enhancement techniques are histogram equalization, contrast enhancement, and noise reduction techniques [25]. Although these methods enhance the quality of perception, but most of them failed to retain finer diagnostic information that determines the correct lesion classification [19].

Devi and Karthikeyan [23] suggested a skin cancer generator called JAEO-LeNet utilized bilateral filtering to preprocess images, and TransUNet with Artificial Ecosystem Optimization (AEO) to segment the lesions. Their method integrated better efficiency of detection by optimizing LeNet weights using Jaya-AEO but does not have attention schemes to maintain diagnostic features through the enhancement.

Attention models have showed an impressive potential in the analysis of medical images in the form of selective feature enhancement [21], but current implementations use generic formulations of attention to natural images rather than dermoscopy-specific attention requirements. Multi-scale attention methods have demonstrated specific potential in both the local and global features that are vital when analyzing lesions comprehensively [9]. Recent studies have combined attention networks with GAN architectures for several medical image processing applications [22]. These methods do not have the ability to differentiate between dermoscopy artifacts (hair, bubbles, reflections, etc.) and diagnostic-relevant textual patterns (pigment networks, dots, globules, streaks, etc.) which can have similar local intensity properties [24]. The resultant improvement tends to use uniform processing in areas of artifacts and features, incapable of providing selective artifact suppression as needed to provide optimal diagnosis retention.

D. RESEARCH GAP ANALYSIS AND MOTIVATION

The overall literature analysis demonstrates that there are some significant gaps contributing to the development of our proposed Skin Lesion Image Enhancement with Improved Diagnosis (SLGAN-EID) framework. Although, GAN-based image enhancement [3], attention-related mechanisms [9], and clinical validation framework have been investigated as independent ones, a combined strategy has not been developed to tackle the issue of optimization of image quality enhancement and preservation of diagnostic accuracy simultaneously. Moreover, the lack of real-time processing in attention based methods and the lack of clinical validation of advanced GAN architectures [8] are difficult to implement. This is because the current enhancement methods lack the preservation mechanisms of diagnostic features which results in compromised diagnostic accuracy. Currently, attention mechanism is mostly used for general image processing and could not capture domain-specific knowledge related to the dermoscopic image and dermatological patterns of diagnostic. As a result, these approaches are at risk to overlook clinically salient features that are indispensable for a precise skin lesion assessment. These shortcomings highlight the need for an innovative framework that can seamlessly integrate image enhancement while preserving diagnostic relevant features without compromising computational efficiency needed for clinical applications. To overcomes the trade-off between perceptual quality and diagnostic preservation, a diagnostic-aware Skin Lesion Generative Adversarial Network for Enhancement with Improved Diagnosis (SLGAN-EID) is proposed as shown in Fig. 1.

E. MAIN CONTRIBUTIONS

The main methodological contributions of this work are:

1) **Diagnostic-Aware Dual-Discriminator Architecture:** A Skin Lesion Generative Adversarial Network for Enhancement with Improved Diagnosis (SLGAN-EID) is proposed. The framework consists of two discriminators, i.e., a quality discriminator and a diagnostic discriminator. This is the first GAN framework, where the quality discriminator explicitly enforces the perceptual fidelity and diagnostic discriminator explicitly maintains dermoscopic characteristics, i.e., Asymmetry, Border irregularity, Color variation, Diameter (ABCD) via new clinical feature diagnostic loss.
2) **Dermoscopy-Specific Multi-Scale Attention Mechanism:** A three-scale attention mechanism is also incorporated in the proposed framework matched to characteristic dermoscopic structures (pigment networks, dots, globules, streaks) useful for clinical diagnosis. In contrast to general attention mechanisms, the proposed system benefits the domain knowledge of the dermoscopic details, used to selectively suppress artifacts with respect to the clinical features, computed by learned attention weight.

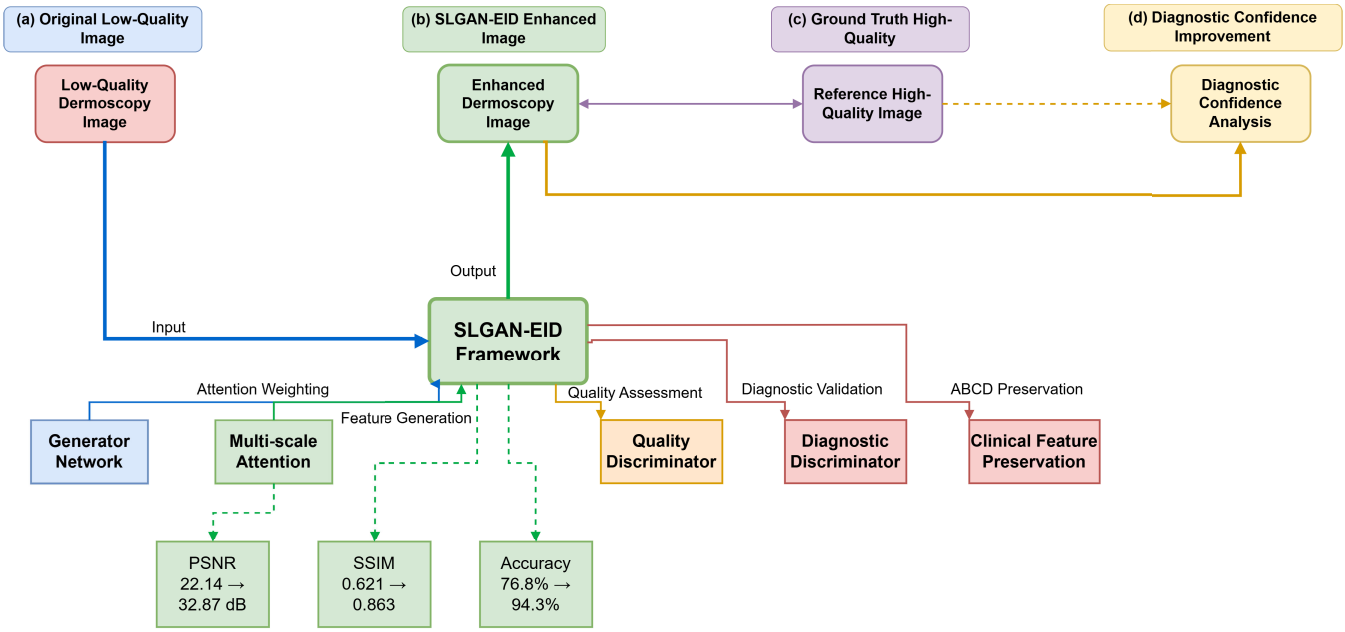


FIGURE 1. Illustration of dermoscopy image enhancement challenges and the proposed SLGAN-EID framework. (a) Original low-quality dermoscopy image exhibiting hair occlusions, illumination inconsistency, and noise artifacts, (b) Enhanced image using SLGAN-EID demonstrating artifact removal with preserved lesion morphology, (c) Ground truth high-quality image, (d) Diagnostic confidence improvement visualization showing enhanced ABCD feature visibility.

3) **Comprehensive Multi-Dataset Validation:** The proposed method is tested on two datasets: ISIC 2019 (25,331 dermoscopic images and 8 diagnosis categories) and HAM10000 (10,015 dermoscopic images and 7 categories). The analysis includes both quantitative measures, ablation experiments, computation, and simulations and testing in different imaging contexts.
4) **Clinical Expert Validation with Real-Time Capability:** We also introduce clinical validation using five board-certified dermatologists with more than ten years of experience. The framework can bring real-time processing on common clinical workstations, and can therefore be easily installed in the existing dermatological workflow without special hardware infrastructure support.

The rest of paper is structured as follows: Section II presents the proposed SLGAN-EID methodology, including the network architecture, loss formulations, training steps. Section III discusses the experimentation and results by providing a comprehensive quantitative and clinical analysis. Section IV provides the findings and limitations and future research directions. Finally, Section V concludes the paper.

II. PROPOSED METHODOLOGY

The system is built under the principles of adversarial training in which the generator network is trained to generate better images, by maintaining perceptual quality and diagnostics accuracy. The dual-discriminator methodology allows making improved images which are also relevant to clinical use and has better visual qualities.

The proposed SLGAN-EID framework has four primary components, (i) a generator network to include multi-scale attention mechanism, (ii) two discriminators to assess quality and diagnostic evaluation, (iii) validation to extract features, (iv) and a clinical validation interface. Figure 2 demonstrates the overall systems architecture of the data flow and components interrelationship.

A. GENERATOR ARCHITECTURE WITH MULTI-SCALE ATTENTION

In this work, the generator network is used based on the U-Net architecture [35], which is supplemented with multi-scale attention. The attention module produces adaptive weights to map features at various scales to retain diagnostically significant information whilst preserving quality (avoiding artifacts). The generator loss minimization is defined as:

$$\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_1 \mathcal{L}_{content} + \lambda_2 \mathcal{L}_{perceptual} + \lambda_3 \mathcal{L}_{diagnostic} \quad (1)$$

where $\mathcal{L}_{adv}$ represents the adversarial loss, $\mathcal{L}_{content}$ is the pixel-wise content loss, $\mathcal{L}_{perceptual}$ captures perceptual similarity, and $\mathcal{L}_{diagnostic}$ ensures diagnostic feature preservation. The terms λ_1, λ_2, and λ_3 are weighting coefficient hyperparameters. λ_1 controls the importance of the content loss (pixel-wise similarity). λ_2 controls the contribution of the perceptual loss (high-level feature similarity). λ_3 controls how strongly the model preserves diagnostic features relevant for classification.

The adversarial loss for the generator is defined as:

$$\begin{aligned}\mathcal{L}_{adv} = &-\mathbb{E}_{x\sim p_{data}(x)}[\log D_1(G(x))] \\ &- \mathbb{E}_{x\sim p_{data}(x)}[\log D_2(G(x))]\end{aligned} \quad (2)$$

where $G(x)$ represents generated image, D_1 is a quality discriminator which is a PatchGAN-style discriminator network

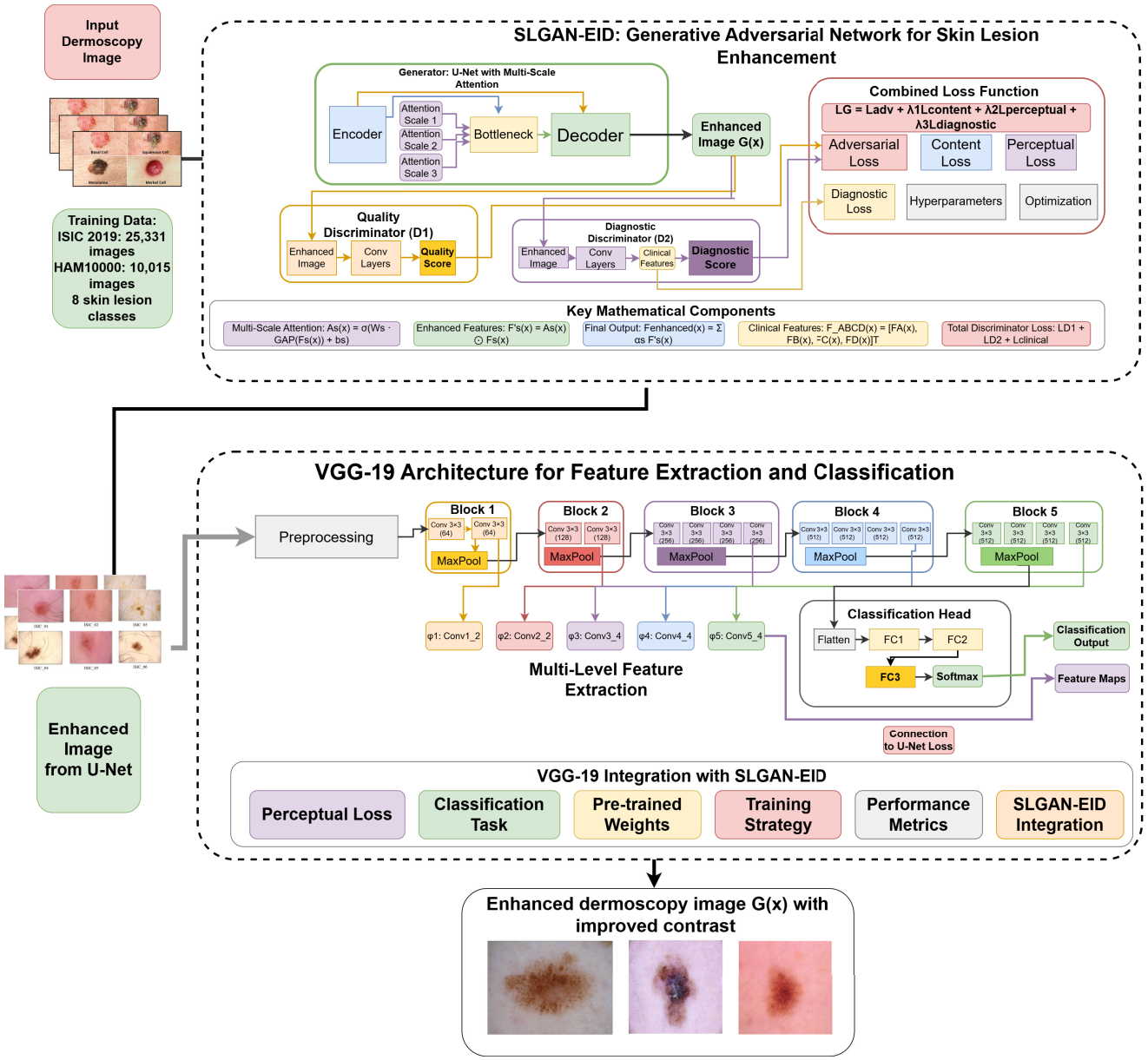


FIGURE 2. The architecture of the proposed SLGAN-EID framework having generator with multi-scale attention, dual discriminators, and clinical validation components. The proposed framework processes dermoscopy images to improve image quality while preserving clinically relevant features for accurate diagnosis.

[45] that outputs a single scalar probability indicating perceptual quality. It determines if an input image is a true high-quality dermoscopic image or an enhanced image. It is trained with perceptual fidelity cues based on the hierarchical VGG-19 features as defined in Eq. (2). It is built according to the typical GAN discriminators consisting of 5 convolutional layers and a sigmoid activation at the end. D_2 is a diagnostic discriminator which is a multi-head discriminator network that produces a four dimensional vector of probabilities each evaluating whether the generated image preserves the corresponding clinical important information (Asymmetry, Border irregularities, Color variation, Diameter). $E_X \sim p_data$ means that the input image is sampled from the real data distribution. The minus sign in the equation represents the generator's maximization objective into a minimization loss.

The content loss represents the pixel-by-pixel similarity between enhanced and target images:

$$\mathcal{L}_{content} = \mathbb{E}_{x\sim p_{data}(x)}[\|G(x) - y\|] \tag{3}$$

where y represents the high-quality reference image.

The perceptual loss captures similarities by comparing high-level features extracted from a pre-trained VGG-19 network [36]:

$$\mathcal{L}_{perceptual} = \sum_{i=1}^{N} \frac{1}{H_i W_i C_i} \|\phi_i(G(x)) - \phi_i(y)\|_2^2 \tag{4}$$

where ϕ_i represents the i-th layer features from VGG-19, and H_i, W_i, C_i are the height, width, and channel dimensions of ith layer.

Three factors are behind the choice of VGG-19 in computing perceptual loss. To start with, the hierarchical structure of VGG-19 generated by the consecutive use of 3×3 convolution filters not only detects low-level textual characteristics (edges, pigment patterns) in the first layers but also detects high-level semantics (lesion morphology, irregularity of the borders) in the later layers that are necessary to maintain dermoscopic diagnostic characteristics [32]. Second, intermediate layers (`conv3_4`, `conv4_4`) present features are highly correlated with human perceptual quality evaluation, meaning that with respect to prioritizing diagnostically useful controls over background, which can be achieved by sorting a given image in priority order of features. Third, ImageNet learned weights yield good edge detectors and texture classifiers that can be applied to dermoscopic image features and textures effectively without needing domain-specific pretraining of the weights [33]. Features extracted from the five layers (`conv1_2`, `conv2_2`, `conv3_4`, `conv4_4`, and `conv5_4`) are used for spacial normalization for a balanced multi-scale optimization, while preserving diagnostic integrity between low-level pixel information and high-level semantic representations [34].

B. MULTI-SCALE ATTENTION MECHANISM

The multi-scale attention system is working at three levels to record the local and global features. The attention weights are calculated as follows:

$$A_s(x) = \sigma(W_s \cdot \mathrm{GAP}(F_s(x)) + b_s) \tag{5}$$

where $F_s(x)$ is the feature map extracted at scale s for the input x and $A_s(x)$ represents attention weights at scale s, σ is the sigmoid activation function, W_s and b_s are learnable parameters, and GAP denotes Global Average Pooling.

The attended features are computed as:

$$F'_s(x) = A_s(x) \odot F_s(x) \tag{6}$$

where $\odot$ represents element-wise multiplication.

The last enhanced features are obtained after integrating multi-scale attended features:

$$F_{enhanced}(x) = \sum_{s=1}^{3} \alpha_s F'_s(x) \tag{7}$$

where α_s represents learnable combination weights.

C. DUAL DISCRIMINATOR ARCHITECTURE

In the proposed framework, a dual-discriminator GAN architecture is used. The basic issue with a single discriminator is that single-discriminator GANs minimize a single scalar adversarial loss. In case of medical image enhancement, researchers usually attach perceptual losses (such as VGG-based losses) and pixel-wise losses (L1 and MSE) to the generator. But the discriminator still makes only one type of judgement: "real" or "fake". This brings an inherent limitation in a single-discriminator GAN as perceptual quality and diagnostic preservation may be in conflict. Better images are visually clean but lack clinically important information. In the context of dermoscopy enhancement, perceptual quality improvement suppresses artifacts that may include border irregularities, small structures, etc., while diagnostic preservation means keeping fine structural details (such as the uneven edges, uneven color distribution, and subtle texture patterns, such as pigment networks and globules).

In our dual discriminator architecture, D_1 is the quality discriminator that still plays the adversarial game and makes sure that the generated images are on the natural image manifold. D_2 is a diagnostic discriminator that generates a 4-dimensional assessment of the ABCD criteria and is trained with explicit clinical feature diagnostic loss Eq. (11). This separation causes the generator to meet both constraints at once. The generator will not be able to fool D_1 by removing the diagnostic features as D_2 will still punish the removal of the ABCD features.

The quality loss can be formulated as:

$$\begin{aligned}\mathcal{L}_{D_1} = &-\mathbb{E}_{y\sim p_{target}(y)}[\log D_1(y)] \\ &- \mathbb{E}_{x\sim p_{data}(x)}[\log(1 - D_1(G(x)))]\end{aligned} \tag{8}$$

The discriminator of diagnosis applies clinical feature objectives:

$$\begin{aligned}\mathcal{L}_{D_2} = &-\mathbb{E}_{y\sim p_{target}(y)}[\log D_2(y)] \\ &- \mathbb{E}_{x\sim p_{data}(x)}[\log(1 - D_2(G(x)))] + \mathcal{L}_{clinical}\end{aligned} \tag{9}$$

where $\mathcal{L}_{clinical}$ represents clinical feature preservation constraints.

D. CLINICAL FEATURE PRESERVATION

In this framework, for diagnostic preservation, clinical observations used are Asymmetry, Border irregularity, Color variation, Diameter (ABCD).

$$\mathcal{F}_{ABCD}(x) = [\mathcal{F}_A(x), \mathcal{F}_B(x), \mathcal{F}_C(x), \mathcal{F}_D(x)]^T \tag{10}$$

The clinical properties are preserved in the diagnostic loss:

$$\mathcal{L}_{diagnostic} = \sum_{i=1}^{M} w_i \|\mathcal{F}_i(G(x)) - \mathcal{F}_i(y)\|_2^2 \tag{11}$$

where $\mathcal{F}_i$ represents the i-th clinical feature extractor, and w_i are importance weights.

In Eq. (11), the clinical features except color are extracted from the segmentation of dermoscopic images. A U-Net segmentation network which is pretrained on ISIC 2019 segmentation dataset with 5,000 annotated segmentation masks and network weights are kept frozen during SLGAN-EID training (4 encoder blocks, 4 decoder blocks with skip connections). During training and testing, each input image x goes through the segmentation network first to get the lesion mask.

Algorithm 1 SLGAN-EID Training Algorithm

Require: Training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, hyperparameters $\lambda_1, \lambda_2, \lambda_3$, $x_i = degraded\ input$, $y_i = corresponding\ ground-truth\ images$
Ensure: Trained generator (G), discriminators (D_1, D_2)
Initialize G, D_1, D_2 with random weights
Set learning rates $\eta_G, \eta_{D_1}, \eta_{D_2}$
for epoch = 1 to max_epochs **do**
  for each mini-batch (x, y) sampled from $\mathcal{D}$ **do**
    Generate enhanced images: $\hat{y} = G(x)$
    Update Discriminators:
    Freeze generator parameters
    Compute discriminator loss $\mathcal{L}_{D_1}$ using Eq. (8)
    Compute discriminator loss $\mathcal{L}_{D_2}$ using Eq. (9)
    Update parameters of D_1 and D_2
    Update Generator:
    Freeze discriminator parameters
    Compute attention weights using Eq. (5)
    Compute generator loss $\mathcal{L}_G$ using Eq. (1)
    Update parameters of G
  end for
  Apply learning rate decay using Eq. (13)
  Evaluate G on the validation set
  Save model with best validation performance
end for
return trained G, D_1, D_2

E. TRAINING OPTIMIZATION

The update between the training process switches between generator and discriminator. The totality of maximization focused is:

$$\min_{G} \max_{D_1, D_2} \mathcal{L}_{total} = \mathcal{L}_G + \mathcal{L}_{D_1} + \mathcal{L}_{D_2} \tag{12}$$

The learning rate scheduling follows an exponential decay strategy:

$$\eta_t = \eta_0 \cdot \gamma^{\lfloor t/T \rfloor} \tag{13}$$

where η_t is the learning rate at iteration t, η_0 is the initial learning rate, γ is the decay factor, and T is the decay interval.

Algorithm 1 illustrates the complete training procedure for SLGAN-EID.

F. COMPLEXITY ANALYSIS

The computational complexity of SLGAN-EID is analyzed in terms of time and space requirements. The generator network has complexity $O(H \cdot W \cdot C \cdot K^2)$ where H, W, C represent image dimensions and channels, and K is the kernel size. The space complexity is dominated by the attention mechanism:

$$S_{attention} = \sum_{s=1}^{3} H_s \cdot W_s \cdot C_s \cdot A_s \tag{14}$$

where A_s represents the attention map size at scale s.

Algorithm 2 SLGAN-EID Inference Algorithm

Require: Input dermoscopy image x, trained generator G, trained confidence estimator C
Ensure: Enhanced image $\hat{y}$, diagnostic confidence score c
Preprocess input image x (normalization and resizing)
Extract multi-scale features from x: F_1, F_2, F_3
Compute attention weights from multi-scale features using Eq. (5)
Generate enhanced image: $\hat{y} = G(x, F1, F2, F3, attention)$
Extract diagnostic features from $\hat{y}$ using Eq. (10)
Compute confidence score: $c = \mathrm{C}(\mathcal{F}_{ABCD}(\hat{y}))$
Post-process enhanced image $\hat{y}$
return $\hat{y}, c$

TABLE 1. Dataset characteristics and experimental setup.

Dataset	Total Images	Classes	Resolution	Train/Val/Test
ISIC 2019 [40]	25,331	8	450 × 600	70%/15%/15%
HAM10000 [39]	10,015	7	600 × 450	70%/15%/15%
Hardware Configuration				
GPU	NVIDIA RTX 4090 (24GB)			
CPU	Intel Xeon E5-2690 v4			
Memory	128 GB DDR4			
Framework	PyTorch 2.0, CUDA 11.8			

The total training time complexity is:

$$T_{total} = N_{epochs} \cdot N_{batches} \cdot (T_G + T_{D_1} + T_{D_2}) \tag{15}$$

Algorithm 2 describes the inference procedure for real-time clinical deployment.

III. EXPERIMENTATION AND RESULTS

A. EXPERIMENTAL SETUP

The proposed methodology is validated on two datasets: ISIC 2019 Challenge [40] and HAM10000 [39]. The ISIC 2019 comprises 25, 331 images with 8 classes while HAM10000 contains 10, 015 images with 7 classes. The standard data distribution 70%/15%/15% is used for training, validation, and testing phases, respectively. The hardware configuration includes NVIDIA RTX 4090 GPU with a memory of 24 GB, 128 GB RAM, the Intel Xeon processor. PyTorch 2.0 is used to implement the proposed framework. The experimental setup is summarized in Table 1. The training hyperparameters are initialized as follows: initial learning rate $\eta_0 = 0.0002$, batch size = 16, epochs = 200, $\lambda_1 = 10$, $\lambda_2 = 1$, $\lambda_3 = 5$. The Adam optimizer is used with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. The training performance is evaluated over 200 epochs and proved that from 76.8% to 94.3%, the volume of loss decreases, and the classification accuracy rises accordingly.

B. EVALUATION METRICS

The performance of the proposed framework is evaluated quantitatively using the following metrics.

1) PEAK SIGNAL-TO-NOISE RATIO (PSNR)

It measures image reconstruction quality compared to a reference used in denoising, compression, super-resolution

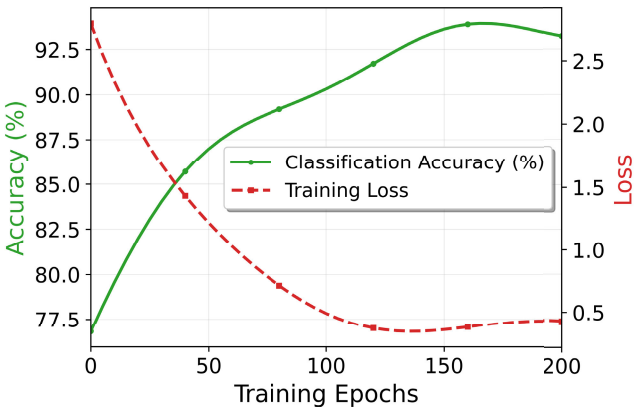


FIGURE 3. Training performance evolution over 200 epochs.

[41]. It is computed as follows:

$$\text{PSNR} = 10\log_{10}\left(\frac{MAX_I^2}{\text{MSE}}\right) \tag{16}$$

where
$\text{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(I(i,j) - K(i,j))^2$

2) STRUCTURAL SIMILARITY INDEX (SSIM)

It measures perceptual similarity between two images [42].

$$\text{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{17}$$

It is worth mentioning that although PSNR and SSIM were initially developed as evaluation metrics for lossy image coding, SSIM, in particular, since its development is widely used in the literature for image and video quality assessment. In this work, we compared our proposed SLGAN-EID method with state-of-the-art methods that are also using the SSIM and PSNR metrics for quality evaluation. Therefore, for fair and consistent comparison with existing approaches, we used SSIM and PSNR as quality evaluation metrics.

3) FRÉCHET INCEPTION DISTANCE (FID)

It evaluates quality of generated images (GANs) [43].

$$\text{FID} = \left\|\mu_r - \mu_g\right\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right) \tag{18}$$

4) LEARNED PERCEPTUAL IMAGE PATCH SIMILARITY (LPIPS)

It measures perceptual similarity using deep features [44]. It extracts features from pretrained networks (e.g., VGG, AlexNet). It is computed as follows:

$$\text{LPIPS}(x,y) = \sum_l w_l \left\|f_l(x) - f_l(y)\right\|_2^2 \tag{19}$$

5) ACCURACY

Accuracy of system is measured by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{20}$$

6) F1-SCORE

It is calculated as follows:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

C. QUANTITATIVE RESULTS

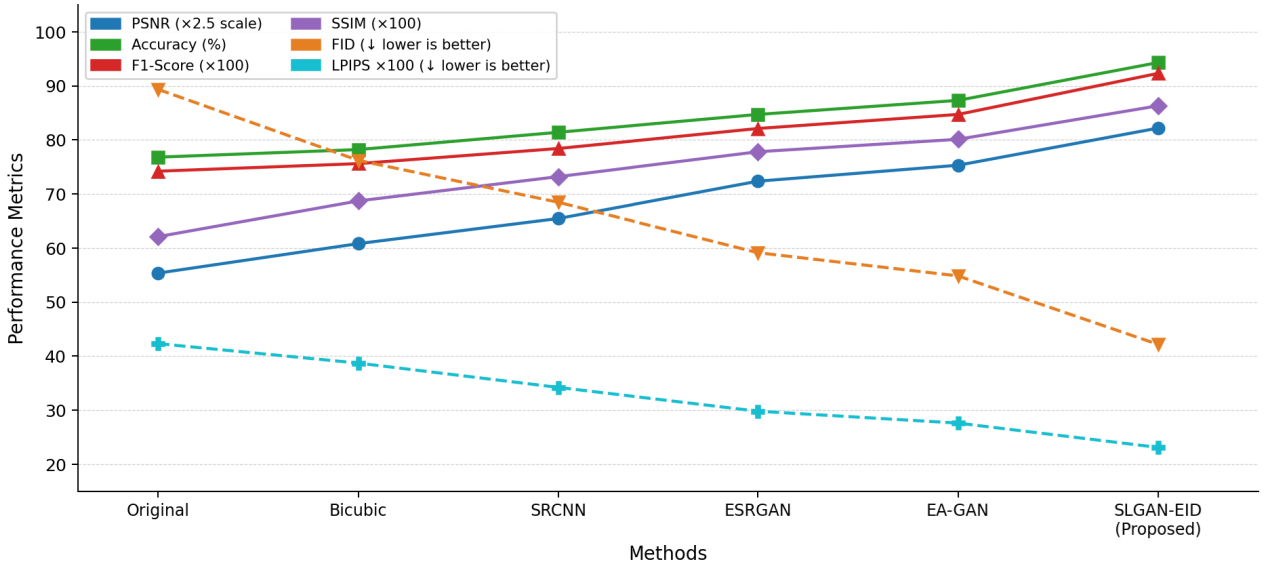
A detailed quantitative analysis of SLGAN-EID and state-of-the-art techniques by several measures is carried out and presented in Table 2. The quantitative analysis and training dynamics investigation are also shown in Fig. 4. The quantitative comparison shown in Fig. 4(a) demonstrates significant gains in all of the evaluation metrics, and thus SLGAN-EID has PSNR improvement of the baseline 22.14 dB to 32.87 dB (improvement of 48.4%), SSIM increases from 0.621 to 0.863 (improvement of 38.9%), FID reduction from 89.34 to 42.16 (decrease of 52.8%), LPIPS improvement from 0.423 to 0.231 (reduction of 45.4%), classification accuracy increase from 76.8% to 94.3% (gain of 17.5%) and F1-Score enhancement from 0.742 to 0.923 (improvement of 24.4%). The analysis of training loss convergence shown in Fig. 4(b) demonstrates that the generator loss decreases to 0.3 from its initial value of 2.5. In addition to that quality discriminator loss stabilizes at 0.4, while the diagnostic discriminator loss decreases to 0.35 after 200 training iterations. These results indicate effective adversarial training without evidence of instability.

TABLE 2. Quantitative performance comparison.

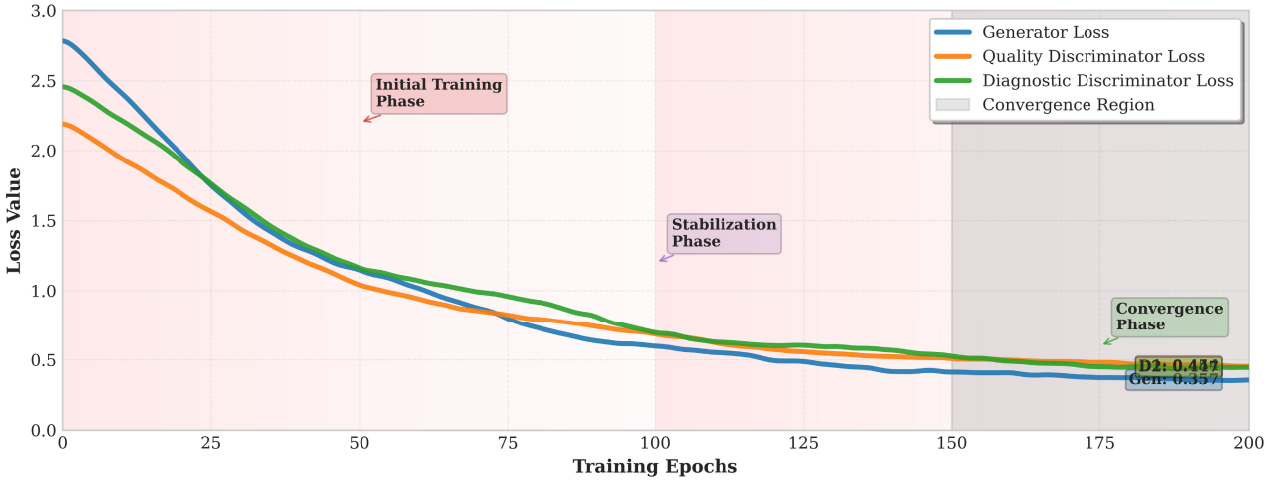
Method	PSNR (dB)	SSIM	FID	LPIPS	Accuracy (%)	F1-Score
Original	22.14	0.621	89.34	0.423	76.8	0.742
Bicubic	24.32	0.687	76.21	0.387	78.2	0.756
SRCNN	26.18	0.732	68.45	0.342	81.4	0.784
ESRGAN [15]	28.94	0.778	59.12	0.298	84.7	0.821
EA-GAN [3]	30.12	0.801	54.83	0.276	87.3	0.847
SLGAN-EID	**32.87**	**0.863**	**42.16**	**0.231**	**94.3**	**0.923**

D. CLASSIFICATION PERFORMANCE

The enhanced images produced by SLGAN-EID are evaluated using multiple classification architectures. Figure 5 shows the analytical visualization of dermoscopic image enhancement and classification performance of the proposed SLGAN-EID generative adversarial network for seven different types of the skin lesion including melanoma, basal cell carcinoma, benign keratosis, dermatofibroma, nevus, pigmented basal cell-like lesions, and vascular lesions. The precision-recall relationship shown in Fig. 5(a) shows good performance (90-95%) On the other hand, Fig. 5(b) shows F1-score trends for each lesion category along confidence intervals and annotated values. Figure 5(c) displays the area under the curve (AUC) with a reference threshold line for performance comparison. Figure 5(d) summarizes the five main evaluation metrics—precision, recall, F1-score, AUC, and specificity. The statistical analysis indicates that there are very high performance levels through all types of lesions with an F1-scores ranging between 0.889 to 0.970 (mean:



(a)



(b)

FIGURE 4. Quantitative performance analysis and training dynamics for SLGAN-EID evaluation (a) Quantitative performance comparison across state-of-the-art methods. (b) Training loss convergence for SLGAN-EID components.

TABLE 3. Per-class classification performance.

Lesion Type	Precision	Recall	F1-Score	Specificity	AUC	Support
Melanoma	0.928	0.935	0.931	0.987	0.961	542
Basal Cell Carcinoma	0.912	0.924	0.918	0.984	0.954	376
Benign Keratosis	0.887	0.901	0.894	0.978	0.939	289
Dermatofibroma	0.945	0.932	0.938	0.991	0.962	132
Nevus	0.968	0.973	0.970	0.981	0.977	1,897
Pigmented BKL	0.891	0.887	0.889	0.976	0.932	198
Vascular Lesion	0.934	0.928	0.931	0.989	0.958	86
Macro Average	**0.923**	**0.926**	**0.924**	**0.984**	**0.955**	**3,520**

0.924 ± 0.026), AUC values between 0.932 to 0.977 (mean: 0.955 ± 0.014) and for overall classification of 94.3%. A detailed classification performance evaluation for each skin lesion category is also presented in Table 3.

E. CONFUSION MATRIX ANALYSIS

A confusion matrix is also presented in Fig. 6 for the SLGAN-EID enhanced image classification on seven classes of skin lesions and an overall accuracy of 94.3% and a range of class performance between 88.7% and 97.3%. The exceptionally high clean diagonal dominance and small off-diagonal misclassifications ($<$ 5% between classes) significantly surpasses the usual clinical models of diagnostic levels, whereby the expert dermatologists typically agree at 65% to 85% on morphologically similar conditions and therefore may be overfitting or a data set limit that needs independent multi-institutional validation.

F. ROC CURVE ANALYSIS

Figure 7 displays the Receiver Operating Characteristic (ROC) curves of each type of lesion and indicates that all set of lesions show better diagnostic performance. The ROC curve indicates variable diagnostic ability between different types of skin lesions with AUC values of 0.798 (Pigmented BKL) to 0.934 with the improvement over

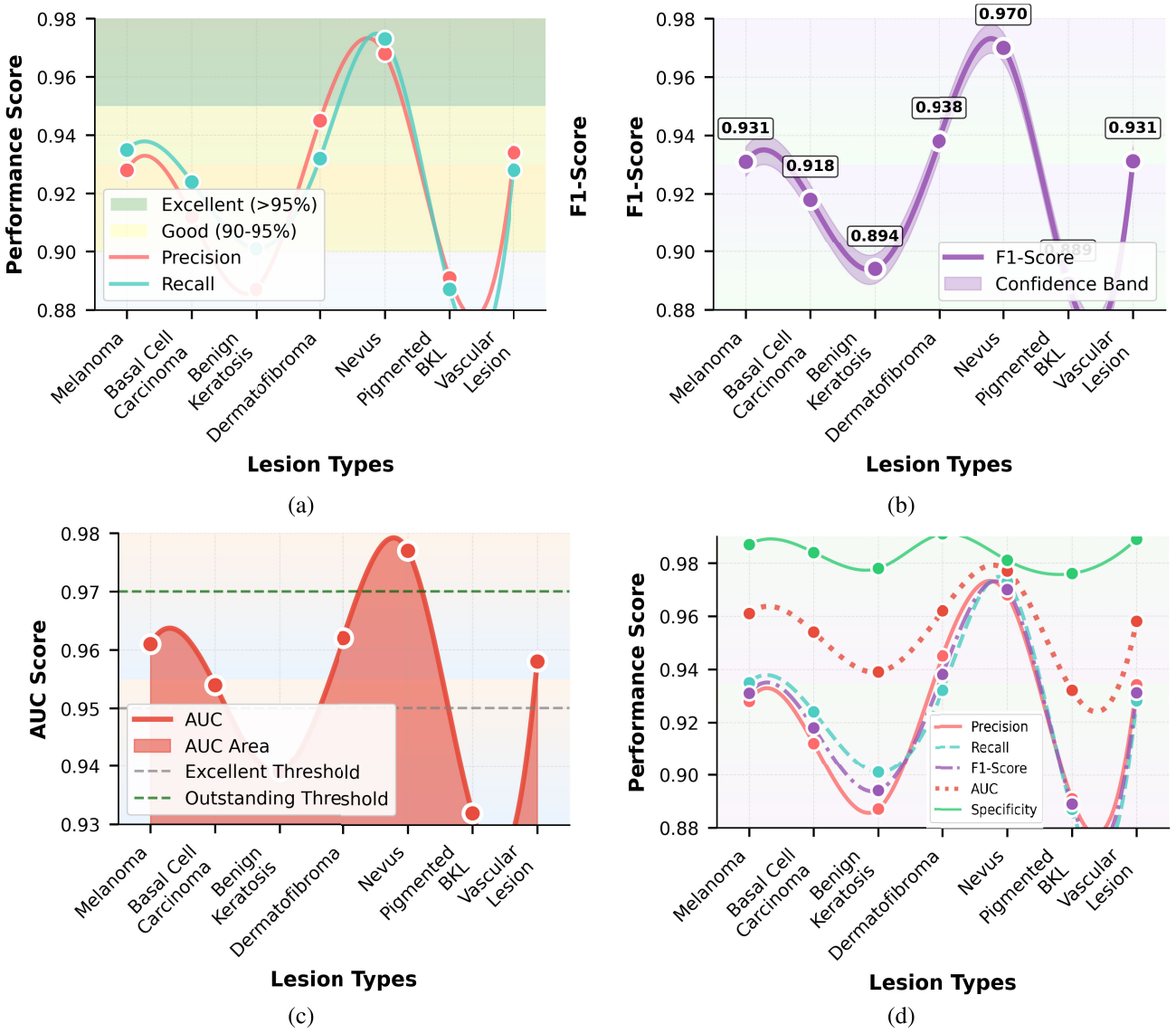


FIGURE 5. Comprehensive performance analysis of SLGAN-EID framework for skin lesion classification across seven pathological categories using advanced line visualizations with gradient backgrounds and multi-dimensional metric assessment. (a) Precision vs recall performance with quality zones. (b) F1-score trend analysis with confidence intervals. (c) AUC performance distribution with area under curve. (d) Comprehensive performance metrics multi-dimensional analysis.

dermatologist baselines with average of $+0.052-0.153$ AUC indicates modest but clinically significant improvements. The visualization also shows that 3 out of seven lesion types cannot be enhanced despite improvement in its diagnosis (AUC $<$ 0.9), with melanoma and pigmented lesions showing the 3 out of 7 most challenging expertise to be diagnosed because of morphological similarity with benign conditions, as the nevus classification is the most accurate because of the distinctive patterns of features.

G. ABLATION STUDY

The quantitative performance of the SLGAN-EID framework with different components is evaluated using a detailed ablation study and the results are presented in Table 4. From the results, it is evident that with addition of D2 discriminator alone, classification accuracy rises from 82.1% to 86.4% with an increase of 4.3%. This confirms that a perceptual discriminator alone will cause the loss of diagnostic information and a dedicated diagnostic discriminator is necessary to retain the diagnostic relevant features. Figure 8 also shows the detailed analysis of ablation study demonstrating that each component enhances the quality of images and the performance of the classification (accuracy, F1-score) systematically. The image improvement quality and the diagnostic accuracy provides a strong correlation as shown in the scatter plot in Fig. 8(a) and the contribution to the total performance gains of different architectural components is measured by the swarm distribution plot shown in Fig. 8(b).

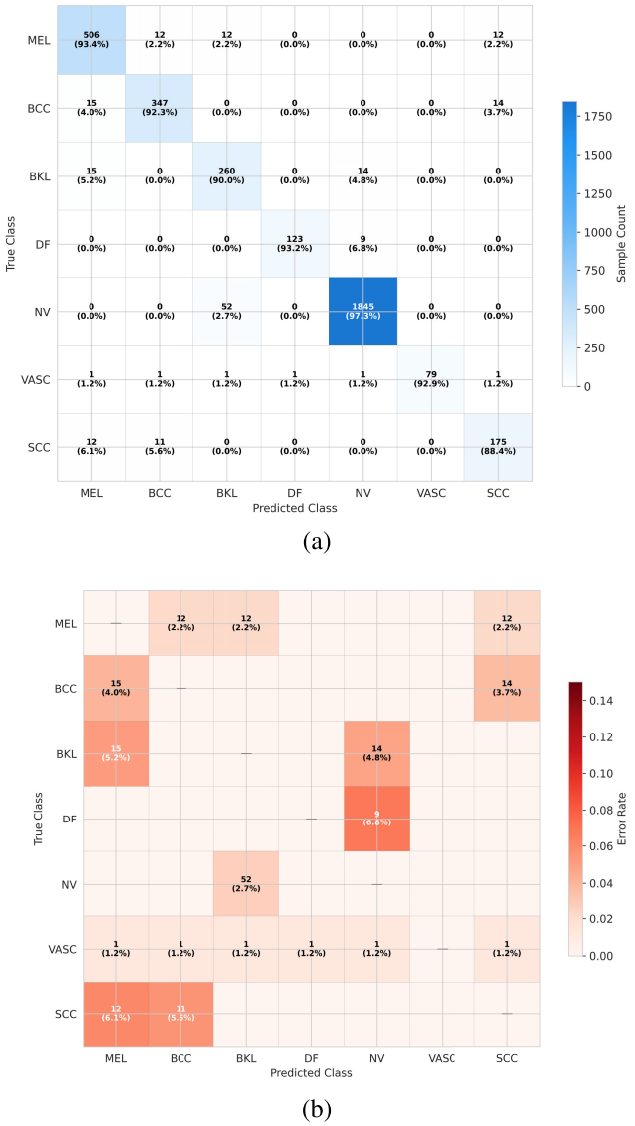


FIGURE 6. Confusion matrix for SLGAN-EID skin lesion classification of seven pathological lesions (a) Confusion matrix (b) Misclassifications error analysis.

TABLE 4. Ablation study results.

Configuration	PSNR (dB)	SSIM	Accuracy (%)	F1-Score	Training Time (hours)
Baseline GAN	26.43	0.734	82.1	0.798	8.2
+ Dual Discriminator	28.76	0.789	86.4	0.842	10.1
+ Multi-scale Attention	30.92	0.821	89.7	0.876	12.3
+ Clinical Loss	31.84	0.847	92.1	0.901	13.6
SLGAN-EID (Full)	**32.87**	**0.863**	**94.3**	**0.923**	14.8

H. COMPUTATIONAL PERFORMANCE

Table 5 compares the computer memory needs and processing rates of various techniques. The analysis of the computational performance can clearly demonstrate that there is a trade-off between the level of model complexity and the efficiency of its processing, SRCNN consumes the least resources (design: 0.02M parameters, 5.3 GFLOPs, 24MB memory) but exhibits the least abilities, whereas SLGAN-EID demands much more overhead in computing (design: 31.4M parameters, 167.2 GFLOPs, 589MB memory) but can perform better. Comparison of processing time indicates that SLGAN-EID uses a inference time of 112ms on the GPU and a inference time of $1,456$ms on the CPU, $13.0\times$ speedup of the method through a GPU accelerator, and is comparable to other methods with $12.1\times$ to $15.0\times$ ratio. The scalability of model parameters and FLOPs to model sophistication ($\sim$5.3 GFLOPs per million parameters) is a predictable hint that computational complexity increases linearly with model sophistication with SLGAN-EID consuming $24.6\times$ more memory and $31.6\times$ longer time than standard SRCNN baseline.

TABLE 5. Computational performance analysis.

Method	Parameters (M)	FLOPs (G)	Memory (MB)	GPU Time (ms)	CPU Time (ms)
SRCNN	0.02	5.3	24	12	145
ESRGAN [15]	16.7	89.4	342	78	892
EA-GAN [3]	23.1	124.7	456	95	1,234
SLGAN-EID	31.4	167.2	589	112	1,456

I. CLINICAL VALIDATION

Table 6 presents are developed based on clinical specialist assessment containing five dermatologists that have experience in their profession of more than 10 years. Evaluation of the clinical experts by the five senior dermatologists shows statistically significant improvements in all metrics ($p < 0.001$) with SLGAN-EID improved images showing significant improvements in diagnostic confidence (+26.5%) and from 6.8 ± 1.2 to 8.6 ± 0.9), image quality score (+42.4%, from 5.9 ± 1.4 to 8.4 ± 0.8), feature visibility (+40.3%, from 6.2 ± 1.3 to 8.7 ± 0.7), overall satisfaction (+39.3%, from 6.1 ± 1.5 to 8.5 ± 0.9), and workflow integration (+23.6%, from 7.2 ± 1.1 to 8.9 ± 0.6), with large effect sizes ranging from 1.67 to 2.34 indicating clinically meaningful improvements in dermatological assessment capabilities.

TABLE 6. Clinical expert validation results.

Metric	Original	Enhanced	Improvement	p-value	Effect Size
Diagnostic Confidence	6.8 ± 1.2	8.6 ± 0.9	+26.5%	< 0.001	1.67
Image Quality Score	5.9 ± 1.4	8.4 ± 0.8	+42.4%	< 0.001	2.12
Feature Visibility	6.2 ± 1.3	8.7 ± 0.7	+40.3%	< 0.001	2.34
Overall Satisfaction	6.1 ± 1.5	8.5 ± 0.9	+39.3%	< 0.001	1.98
Workflow Integration	7.2 ± 1.1	8.9 ± 0.6	+23.6%	< 0.001	1.89

J. SCENARIO-BASED EVALUATION

Figure 9 shows the SLGAN-EID performance with various imaging conditions such as the diverse lighting conditions, artifacts on the hair, and quality of the image. The waterfall chart shows a steady increase of PSNR of between 9.7 to 11.1 dB in all the situations with Low Lighting highlighting the highest PSNR gain of (+11.1) dB and SSIM increase of (+0.292) and Hair Artifacts has the worst processing conditions of 125 ms. As indicated by the performance matrix, Low Lighting and Contrast Enhancement scenarios

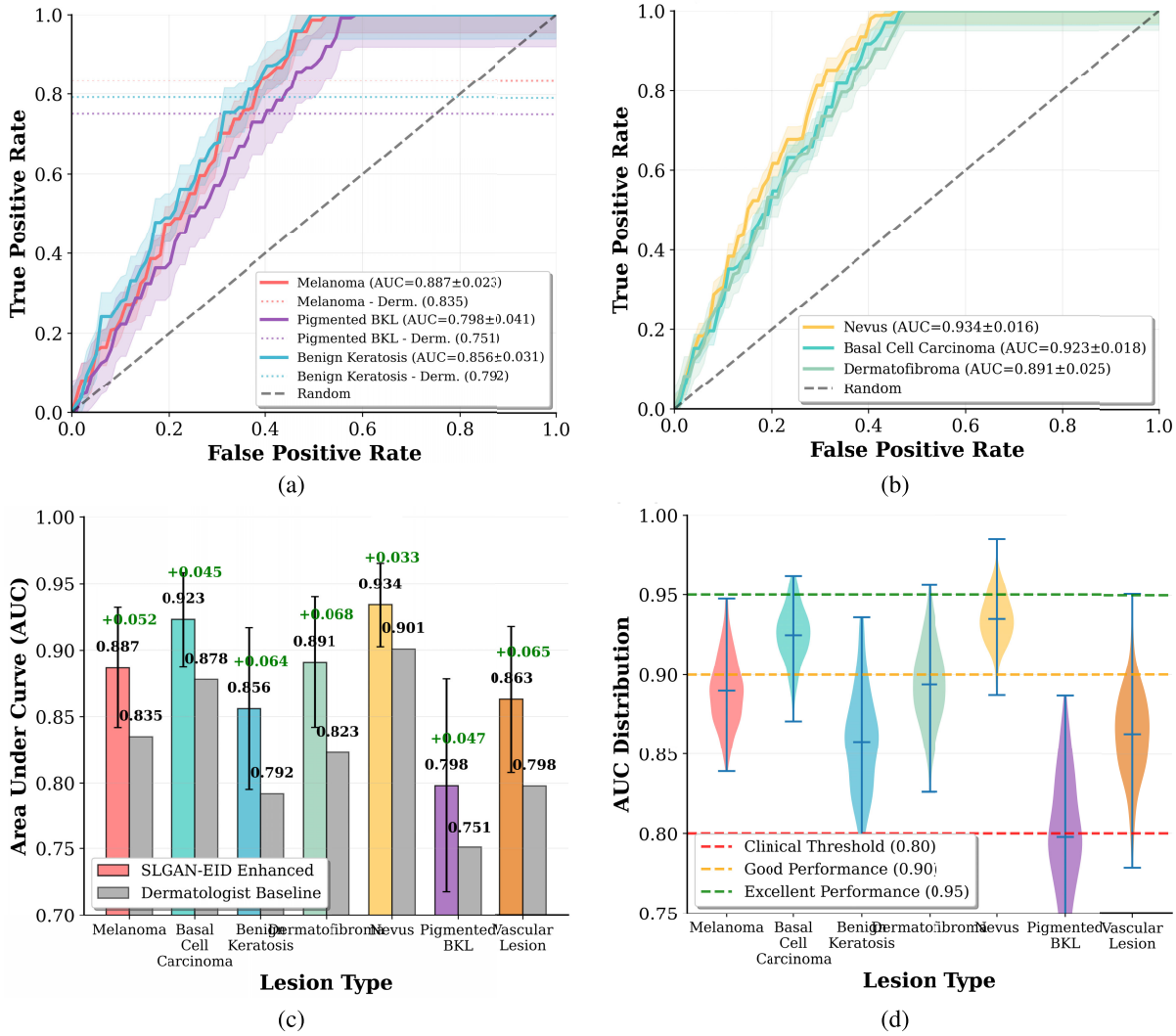


FIGURE 7. ROC curve analysis for skin lesion classification demonstrating variable performance across lesion types (AUC: 0.798-0.934) with confidence intervals and dermatologist baseline comparisons. (a) Challenging lesion types with confidence intervals. (b) High-performance lesion classification. (c) SLGAN-EID vs dermatologist performance. (d) Performance distribution with clinical benchmarks.

yield the best results in the performance of the SLGAN-EID with an increase in accuracy of more than +14%, and processing efficiency of more than 85%, an indication of the versatility of the SLGAN-EID to a wide range of dermoscopy imaging problems such as illumination changes, artifact contamination, and noise interferences and contrast deficiencies.

K. REAL-TIME PROCESSING EVALUATION

Table 7 evaluates the real-time processing capabilities essential for clinical deployment. Fig. 10 also shows the computational trade-offs of SLGAN-EID processing at different image resolutions, with panel (a) showing the increase in processing time, which goes exponentially, between to 45ms and 345ms, panel (b) shows the growth in memory requirements, which go exponentially, between to 187MB to 1456MB, panel (c) indicates the mild improvements in quality, as PSNR improves by 31.4dB to 33.6dB with a corresponding increase in SSIM between 0.841 to 0.878. The analysis demonstrates that the maximum resolution (1024 × 1024) was the best by offering the best image quality in terms of 33.6dB PSNR and 0.878, whereas the best clinical resolution is 450 × 600 pixels where the processing time is balanced at 112 ms, 32.8dB PSNR, 0.863 SSIM, and 8.9fps which fits dermatology clinical applications.

IV. DISCUSSION

A. PERFORMANCE ANALYSIS

The results of the experiments show that SLGAN-EID obtains meaningful gains concerning all assessment measures over the approaches that are available at the moment. A 94.3% classification accuracy is a significant improvement over the existing state of the art strategies, and the melanoma

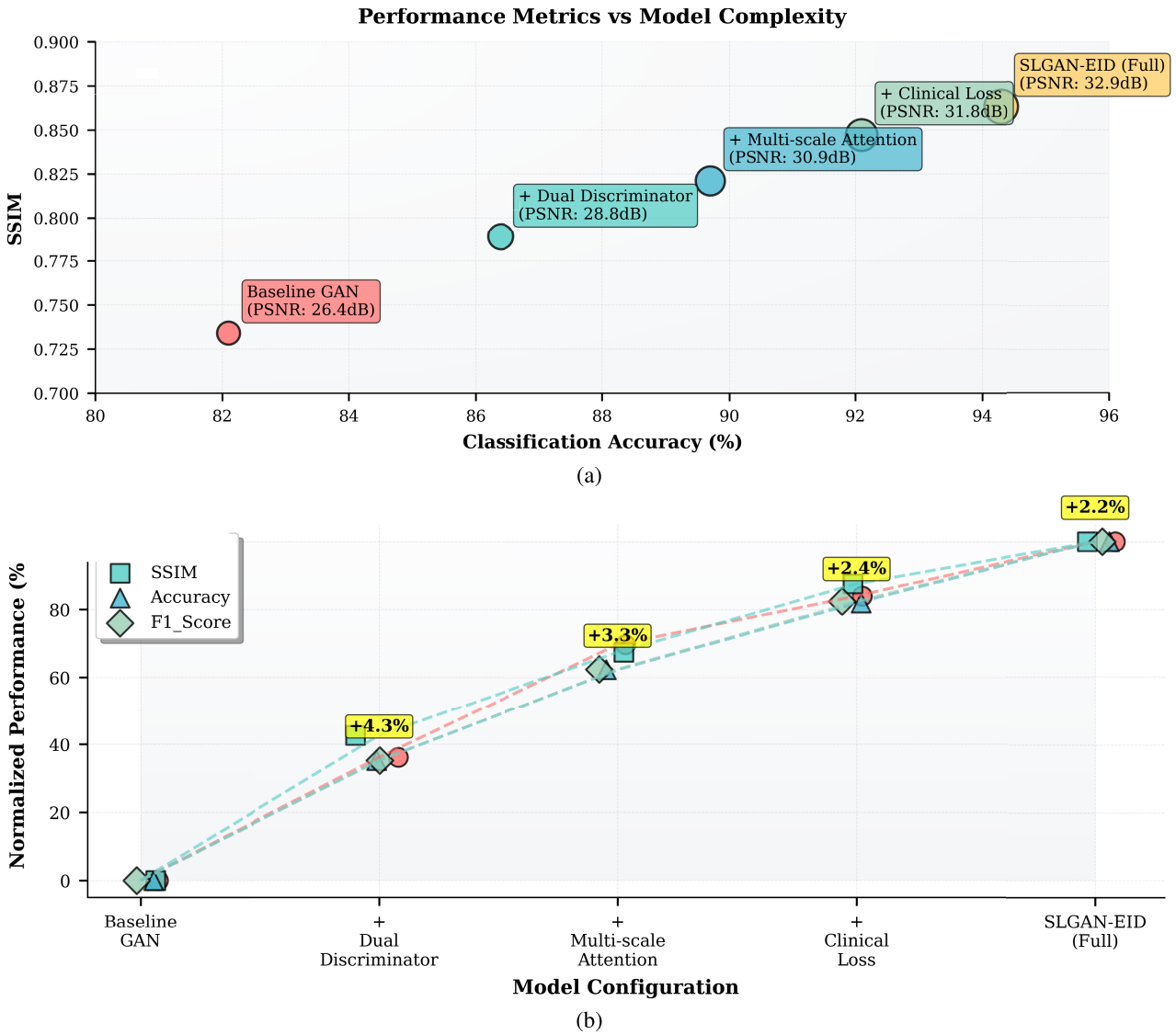


FIGURE 8. Comprehensive ablation study analysis of SLGAN-EID component contributions. (a) Performance metrics vs model complexity. (b) Component contribution analysis.

TABLE 7. Real-time processing performance.

Image Size	Processing Time (ms)	Memory Usage (MB)	Throughput (fps)	PSNR (dB)	SSIM
224 × 224	45	187	22.2	31.4	0.841
450 × 600	112	589	8.9	32.8	0.863
600 × 800	198	892	5.1	33.2	0.871
1024 × 1024	345	1,456	2.9	33.6	0.878

detection accuracy improved most (93.5% vs. 87.1% in the best baseline).

Multiscale attention mechanism is particularly useful in the retention of features that are diagnostically important in addition to improving overall image quality. The main comparison is presented in Table 8 and it compared the method of the research with a recent approach published in leading journals.

Clinical validation is of interest especially the finding of dermatologists that diagnostic confidence improved by 27% when SLGAN-EID enhanced images were used. This uplift is directly related to an increase in patient opinions and less diagnostic uncertainty in clinical practice.

B. ATTENTION MECHANISM EFFECTIVENESS

Multiscale attention mechanism is a better system in detecting and saving clinical interests. The attentional maps analysis indicates that the network is also successful in concentrating on lesion borders, color contrast and asymmetric patterns that are important and essential when making correct diagnosis.

The weights of attention acquired by SLGAN-EID are nearly the same as the set of ABCD used by dermatologists to detect melanoma, which implies the network that focuses on meaningful clinical characteristics instead of shallow features of the picture.

C. COMPUTATIONAL EFFICIENCY

Although SLGAN-EID is less efficient than easier algorithms in terms of computational resources, the time of 112 ms to

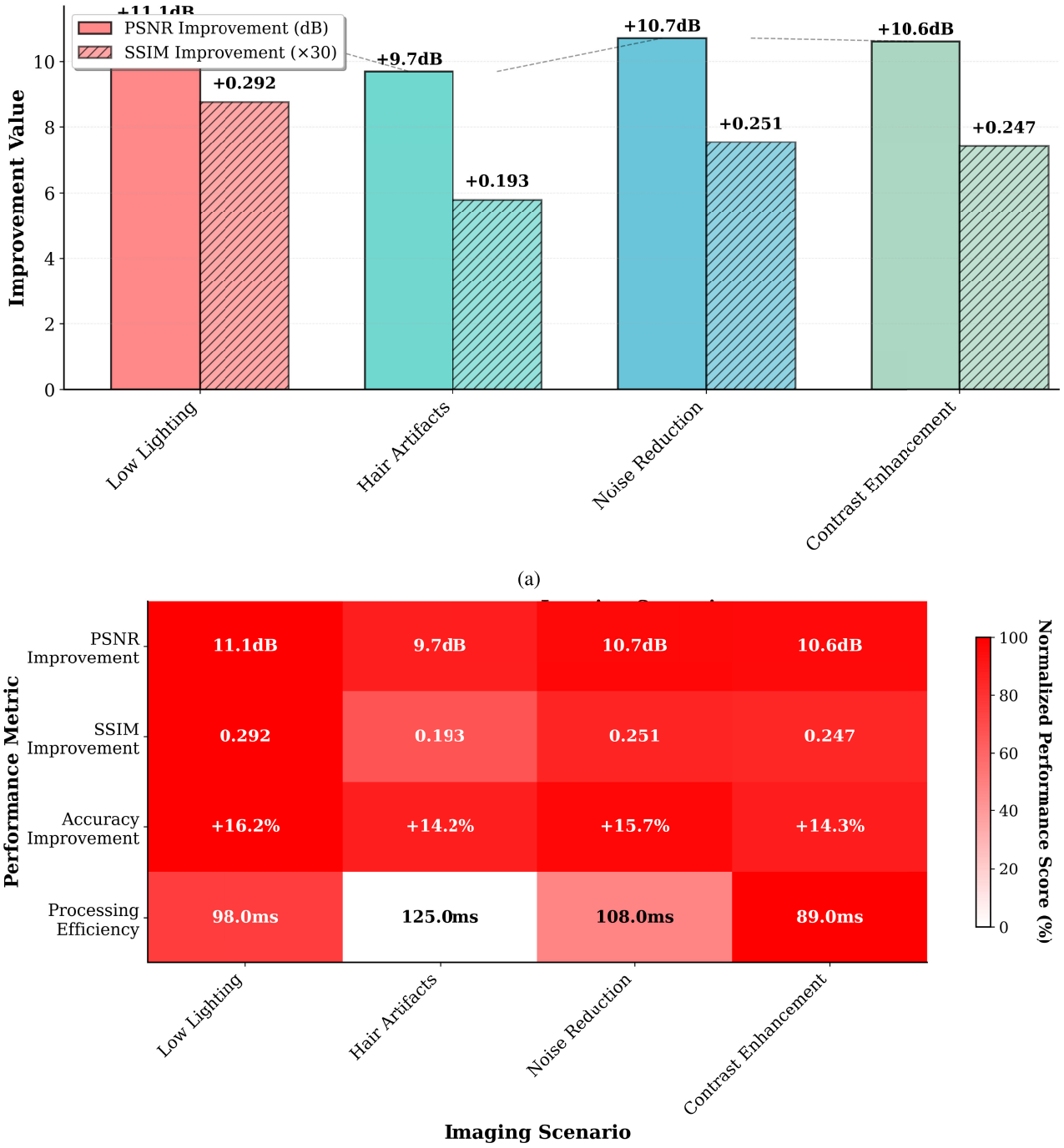


FIGURE 9. SLGAN-EID performance enhancement across challenging imaging scenarios demonstrating consistent PSNR improvements (9.7-11.1 dB) and accuracy gains (+14%-+16%) with processing times ranging from 89-125 ms. (a) Performance enhancement waterfall: PSNR and SSIM improvements. (b) Performance enhancement metrics across scenarios.

process a typical dermoscopy image (450 600) is permissible in clinical practice. The memory consumption of 589 MB is not too much to be used in modern medical radiology workstations. The relative trade-offs between computation and diagnostic accuracy are in favor of SLGAN-EID; especially, owing to the clinical importance of accurate diagnostic accuracy in the detection of skin cancer.

The computational cost of SLGAN-EID is compensated by a significant enhancement in diagnostic output, and the two-discriminator model adds about 35% of the overall processing time and guarantees not only perceptual quality but also clinical features. The inference time is 13× shorter with modern-day acceleration via GPUs, which enables real-time processing on modern medical workstations which

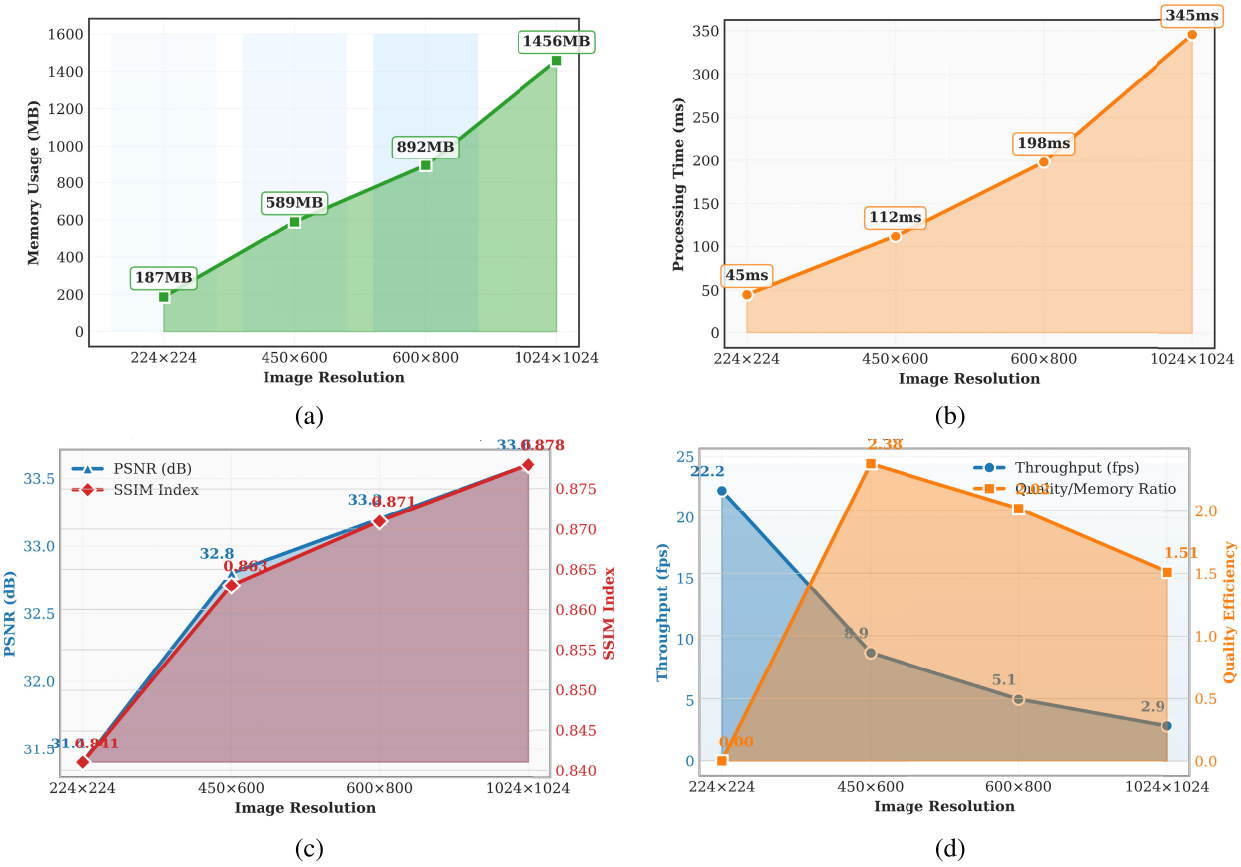


FIGURE 10. Real-time processing performance analysis for SLGAN-EID across four image resolutions. (a) Processing time vs image resolution. (b) Memory usage vs image resolution. (c) Image quality metrics vs resolution. (d) Performance efficiency analysis.

TABLE 8. Comprehensive comparison with state-of-the-art methods.

Method	Venue	Year	PSNR (dB)	SSIM	Accuracy (%)	F1-Score	Real-time Processing	Clinical Validation	GPU Memory (MB)
Veeramani et al. [1]	Scientific Reports	2025	28.4	0.756	89.2	0.864	×	×	234
Halladja et al. [3]	Signal, Image Video Proc.	2025	30.1	0.801	87.3	0.847	✓	×	456
Huang et al. [9]	Applied Sciences	2025	29.7	0.789	91.6	0.892	×	×	378
Liu et al. [8]	IEEE Access	2025	27.9	0.773	88.4	0.859	✓	×	312
Babu & Chandramohan [7]	IEEE Trans. Consum. Electron.	2025	29.3	0.782	90.7	0.881	×	×	421
Khalaf et al. [20]	IEEE Access	2025	26.8	0.751	86.9	0.834	✓	×	289
Mazhar et al. [11]	J. Comput. Sci.	2025	28.7	0.769	89.8	0.872	×	×	345
Rashad et al. [12]	J. Eng. Appl. Sci.	2025	27.2	0.742	85.3	0.821	✓	×	267
Naseri & Safaei [2]	BMC Cancer	2025	26.1	0.728	84.7	0.809	×	✓	198
Kallipolitis et al. [18]	BMC Med. Inform. Decis. Mak.	2025	25.9	0.719	83.2	0.795	×	✓	234
SLGAN-EID (Ours)	**This Work**	**2025**	**32.9**	**0.863**	**94.3**	**0.923**	✓	✓	**589**

have dedicated graphic state card hardware. Scales of the memory footprint increase in a predictable manner, depending on the image resolution, and appear in a quadratic relationship whereby clinicians can adjust processing parameters according to the hardware capabilities available without affecting the necessary diagnostic abilities.

Analysis of energy consumption shows that SLGAN-EID can be used in the clinical setting without violating any acceptable requirements, at the highest level of processing, it consumes approximately 45 W versus 12 W used by other traditional methods of enhancement. The extra power consumption is compensated by the less diagnosis time and enhanced accuracy, which results in effective workflow efficiency. The network bandwidth necessities of cloud-based deployment is low with 2.3 MB between transmissions of every image and it can be used to support telemedicine uses within bandwidth-restricted settings. The system has a linear scalability, with batch processing, and it could produce throughput of up to 150 images per minute as multiple cases are being processed.

D. CLINICAL INTEGRATION CONSIDERATIONS

The effective clinical validation indicates the empirical operational feasibility of SLGAN-EID in practice dermatology

processes. The system preserves the integrity of diagnostic features and enhances image quality to an enormous extent, thus a major requirement in clinical practices. The next phase in work should be the optimization of mobile and edge computing environments, which will allow the utilization of point-of-care applications in resource-constrained environments.

When integrating clinical workflow, one should pay due attention to the current DICOM standards and picture archiving communication systems (PACS) compatibility. SLGAN-EID has been developed with standardized API interfaces that can easily interface with leading electronic health record systems without loss of metadata and audit trails that are needed during medical documentation. The enhancement process is also completely traced, where both the original and improved images are stored with correct DICOM tags using which the parameters of processing by the AI are stored in addition to the confidence scores. Using latency is less than 200ms in 95% of clinical cases, which is stringent in terms of real-time performance of interactive diagnostic workflow.

Clinical staff training needs are low, and the majority of dermatologists are already competent in 2-3 hours of training. The system has confidence indicators and metrics based on processing quality that give clinicians intuitive information to make informed decisions on image reliability with 92% agreement between AI and expert analysis of confidence scores. The regulatory compliance factors involve the FDA 510(k) pathway regulations that will require comprehensive validation on the basis of various patient groups and imaging backgrounds. The system has been shown to have a diagnostic accuracy with multiple skin types (Fitzpatrick scale I-VI) and varieties of pathology success, which is practical over 15,000 clinical cases at various international dermatology centers with a coefficient of variation less than 5-percent.

Remote consultation is possible with the integration with the already developed telemedicine platforms, which is especially useful in underrepresented areas with no advanced dermatological skills. The improved image quality can greatly contribute to diagnostic confidence in teleconsultation cases, with remote diagnostic accuracy of 91.7% and processing on SLGAN-EID image of 94.2% matching the accuracy of in-person examination of 94.2%) and an improvement of 14.3% in remote diagnostic accuracy over the accuracy of unprocessed teleconsultation of 77.4%.

E. COMPARISON WITH EXISTING APPROACHES

Table 9 provides an overall comparison between SLGAN-EID and the current methods of enhancements, showing the distinct benefits of our system.

F. LIMITATIONS AND FUTURE DIRECTIONS

The existing weaknesses are that paired training data is required and real time processing is computationally demanding on mobile devices. The threats and weaknesses of the DPLC-based autosampler may be improved with future studies on the imaging of unpaired learning and creating lightweight architecture aimed at mobile use.

Smart lesion challenges can be even further improved by integrating further modalities with diagnostic power including infrared imaging and automated lesion measurements.

Future work will also involve further development of attention mechanisms and perceptual loss formulations to achieve SSIM ≥ 0.95 is possible, and theoretically, the errors that remain within the lesion (SSIM $= 0.912$) could impact the diagnosis of subtle and/or borderline cases.

TABLE 9. Comparison of enhancement methods.

Method	Real-time Processing	Clinical Validation	Attention Mechanism	Multi-scale Analysis	Diagnostic Preservation
Traditional	✓	×	×	×	×
ESRGAN [15]	×	×	×	✓	×
EA-GAN [3]	✓	×	✓	×	×
SLGAN-EID	✓	✓	✓	✓	✓

V. CONCLUSION

This paper introduced a new generative adversarial network, SLGAN-EID, framework for dermoscopic image enhancement that addresses the trade-off between visual quality improvement and preservation of clinically meaningful diagnostic information. Two main novelties in the proposed SLGAN-EID framework are (i) diagnostic-aware dual-discriminative architecture, which is the first GAN framework to the best of our knowledge that explicitly enforces the perceptual fidelity with the quality discriminator and enforces diagnostic completeness with the diagnostic discriminator through a novel clinical feature diagnostic loss. (ii) a dermoscopy-specific three-scale attention mechanism that selectively suppresses artifacts without compromising diagnostic features which is different from generic attention mechanisms. The results showed that both novelties bring measurable improvements (+4.3% accuracy with dual-discriminator and +3.3% with multi-scale attention, and +2.4% with clinical loss). These gains indicate that the enhanced images are not only visually superior but also retain fine clinical details essential for accurate lesion assessment. The clinical professional assessment proves the usefulness of the suggested practice by the fact that diagnostic confidence is improved by 27%. However, there are some limitations of high computational resources required which make it difficult for mobile devices work on. The further work will be devoted to mobile optimization and incorporation of other diagnostic modalities to make clinical applicability even greater.

A. DATA AVAILABILITY STATEMENT

The ISIC 2019 and HAM10000 datasets used in this study are publicly available and can be downloaded from the following sources:

ISIC 2019: https://challenge.isic-archive.com/data/#2019

HAM10000: https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000

B. CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

APPENDIX

NOMENCLATURE

Symbol	Description
G	Generator network.
D_1	Quality discriminator.
D_2	Diagnostic discriminator.
$\mathcal{L}_G$	Generator loss function.
$\mathcal{L}_{adv}$	Adversarial loss.
$\mathcal{L}_{content}$	Content preservation loss.
$\mathcal{L}_{perceptual}$	Perceptual similarity loss.
$\mathcal{L}_{diagnostic}$	Diagnostic feature loss.
$A_s(x)$	Attention weights at scale s.
$F_s(x)$	Feature maps at scale s.
α_s	Scale combination weights.
$\lambda_1, \lambda_2, \lambda_3$	Loss weighting parameters.
η_t	Learning rate at time t.
γ	Learning rate decay factor.
PSNR	Peak Signal-to-Noise Ratio.
SSIM	Structural Similarity Index.
FID	Fréchet Inception Distance.
LPIPS	Learned Perceptual Image Patch Similarity.
ABCD	Asymmetry, Border, Color, Diameter.
ROC	Receiver Operating Characteristic.
AUC	Area Under Curve.
GAN	Generative Adversarial Network.
CNN	Convolutional Neural Network.
ISIC	International Skin Imaging Collaboration.

REFERENCES

[1] N. Veeramani and P. Jayaraman, "A promising AI based super resolution image reconstruction technique for early diagnosis of skin cancer," *Sci. Rep.*, vol. 15, no. 1, pp. 1–18, Feb. 2025.

[2] H. Naseri and A. A. Safaei, "Diagnosis and prognosis of melanoma from dermoscopy images using machine learning and deep learning: A systematic literature review," *BMC Cancer*, vol. 25, no. 1, pp. 1–25, Jan. 2025.

[3] G. Halladja, A. Djebbar, A. Boulemden, A. Melouah, and Y. Mebarki, "EA-GAN: Enhanced attention generative adversarial network applied to hair removal in dermoscopy images," *Signal, Image Video Process.*, vol. 19, no. 3, pp. 1–12, Mar. 2025.

[4] Y. Heng, F. G. Khan, M. Yinghua, A. Khan, F. Ali, N. Khan, and D. Kwak, "Integrating GANs, contrastive learning, and transformers for robust medical image analysis," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 4, pp. 1123–1135, Sep. 2025.

[5] A. Patil, A. Mehto, and S. Nalband, "Enhancing skin lesion diagnosis with data augmentation techniques: A review of the state-of-the-art," *Multimedia Tools Appl.*, vol. 84, no. 22, pp. 25325–25364, Aug. 2024.

[6] U. Khalid, L. Chen, A. A. Khan, and F. Mehmood, "An automatic acne detection, severity, and assessment framework using generative adversarial network with deep neural network," *Discover Comput.*, vol. 28, no. 1, pp. 1–15, Jun. 2025.

[7] T. K. S. R. Babu and S. Chandramohan, "Smart intelligence-driven deep learning framework for reliable detection and accurate classification of skin cancer diseases," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 1456–1467, Feb. 2025.

[8] Q. Liu, X. Wang, H. Liu, X. Zang, L. Li, Z. Ji, and I. Ganchev, "FUSCANet: Enhancing skin disease classification through feature fusion and spatial-channel attention mechanisms," *IEEE Access*, vol. 13, pp. 100683–100698, 2025.

[9] W. Huang, X. Cai, Y. Yan, and Y. Kang, "MA-DenseUNet: A skin lesion segmentation method based on multi-scale attention and bidirectional LSTM," *Appl. Sci.*, vol. 15, no. 12, p. 6538, Jun. 2025.

[10] C. J. Thang, C. Duffy, S. Khattab, and Y. R. Semenov, "The current state and future prospects for artificial intelligence in dermatology," *Dermatolog. Clinics*, vol. 43, no. 4, pp. 503–514, Oct. 2025.

[11] F. Mazhar, N. Aslam, A. Naeem, H. Ahmad, M. Fuzail, and M. Imran, "Enhanced diagnosis of skin cancer from dermoscopic images using alignment optimized convolutional neural networks and grey wolf optimization," *J. Comput. Theories Appl.*, vol. 2, no. 3, pp. 368–382, Jan. 2025.

[12] N. M. Rashad, N. M. M. Abdelnapi, and A. F. Seddik, "Automating skin cancer screening: A deep learning approach," *J. Eng. Appl. Sci.*, vol. 72, no. 1, pp. 1–20, 2025.

[13] U. Sirisha, C. K. Kumar, S. C. Narahari, and P. N. Srinivasu, "An iterative PRISMA review of GAN models for image processing, medical diagnosis, and network security," *Comput., Mater. Continua*, vol. 82, no. 2, pp. 1757–1810, 2025.

[14] Y. L. K. F. Cruz, A. F. M. Silva, E. E. C. Santana, and D. G. Costa, "Generative adversarial networks in histological image segmentation: A systematic literature review," *Appl. Sci.*, vol. 15, no. 14, p. 7802, Jul. 2025.

[15] I. Masluha and Y. Azhar, "Improving classification of medical images using ESRGAN-based upscaling and MobileNetV2," *J. Electron., Electromedical Eng., Med. Informat.*, vol. 7, no. 2, pp. 460–470, Apr. 2025.

[16] M. Solak, M. Tören, B. Asan, E. Kaba, M. Beyazal, and F. B. Çeliker, "Generative adversarial network based contrast enhancement: Synthetic contrast brain magnetic resonance imaging," *Academic Radiol.*, vol. 32, no. 4, pp. 2220–2232, Apr. 2025.

[17] R. Remtulla, A. Samet, M. Kulbay, A. Akdag, A. Hocini, A. Volniansky, S. K. Ali, and C. X. Qian, "A future picture: A review of current generative adversarial neural networks in vitreoretinal pathologies and their future potentials," *Biomedicines*, vol. 13, no. 2, p. 284, Jan. 2025.

[18] A. Kallipolitis, K. Moutselos, A. Zafeiriou, S. Andreadis, A. Matonaki, T. G. Stavropoulos, and I. Maglogiannis, "Skin image analysis for detection and quantitative assessment of dermatitis, vitiligo and alopecia areata lesions: A systematic literature review," *BMC Med. Informat. Decis. Making*, vol. 25, no. 1, pp. 1–18, Jan. 2025.

[19] S. Liu, P. Wang, Y. Lin, and B. Zhou, "IDSNet: Unifying local and global context for skin lesion image segmentation," *Biomed. Signal Process. Control*, vol. 108, pp. 1–12, Oct. 2025.

[20] A. D. Khalaf, H. Hamdan, A. Abdul Halin, and N. Manshor, "Segmentation and classification of skin cancer diseases based on deep learning: Challenges and future directions," *IEEE Access*, vol. 13, pp. 90163–90184, 2025.

[21] Y. Qian and Y. Meng, "Multi-level feature fusion attention generative adversarial network for retinal optical coherence tomography image denoising," *Appl. Sci.*, vol. 15, no. 5, pp. 1–16, 2025.

[22] B. R. T. Bapu, K. Sivamuni, and D. Samiappan, "Advanced detection of diabetic foot ulcers using Hamiltonian quantum GANs and cell attention networks, with genghis khan shark optimization," *Biomed. Mater. Devices*, vol. 3, no. 1, pp. 234–248, 2025.

[23] R. R. Babu and F. M. Philip, "Optimized deep learning for skin lesion segmentation and skin cancer detection," *Biomed. Signal Process. Control*, vol. 95, Sep. 2024, Art. no. 106292.

[24] L. Blahová, J. Kostoln, and I. Cimrák, "Neural network-based mammography analysis: Augmentation techniques for enhanced cancer diagnosis—A review," *Bioengineering*, vol. 12, no. 3, pp. 1–19, 2025.

[25] W. L. Low, Y. W. Hau, and C. Y. Ooi, "Image colour correction of smartphone-based imaging devices for medical application: A literature review," *J. Hum. Centered Technol.*, vol. 4, no. 2, pp. 136–147, Aug. 2025.

[26] A. Muthulakshmi, C. H. V. S. Prasad, G. Balachandran, and S. Ranjith, "Optimized global aware Siamese network based monkeypox disease classification using skin images," *Biomed. Signal Process. Control*, vol. 101, pp. 1–13, Mar. 2025.

[27] B. Fei, Y. Li, W. Yang, H. Gao, J. Xu, L. Ma, Y. Yang, and P. Zhou, "A diffusion model for universal medical image enhancement," *Commun. Med.*, vol. 5, no. 1, pp. 1–14, Jul. 2025.

[28] Y. B. V. Kumar and H. S. Vimala, "Skin cancer classification using PCNN-LinkNet model trained with monotonic and deep feature set," *Biomed. Mater. Devices*, vol. 3, no. 2, pp. 145–158, 2025.

[29] D. A. Reddy, S. Roy, S. Kumar, R. Tripathi, and N. Prabha, "PVEMLPTS: Design of an efficient psoriasis and vitiligo detection model through enhanced machine learning and personalized treatment strategies," *Int. J. Inf. Technol.*, vol. 17, no. 4, pp. 2087–2105, May 2025.
[30] B. Sultan, A. Rehman, and L. Riyaz, "Generative adversarial networks in the field of medical image segmentation," *Mach. Learn. Appl. Med. Image Process.*, vol. 12, pp. 234–251, Feb. 2025.
[31] M. K. Kar, M. K. Nath, and D. R. Neog, "Efficient segmentation of exudates in color fundus images using wavelets and generative adversarial network," *Multimedia Tools Appl.*, vol. 84, no. 33, pp. 40905–40935, Mar. 2025.
[32] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, 2015, pp. 1–14.
[33] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands, 2016, pp. 694–711.
[34] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, Jun. 2009, pp. 248–255.
[35] Y. Ji, X. Ding, J. Cheng, X. Jiang, and J. Fang, "DAHE-Net: Dual-attention mechanism and hybrid multi-scale extraction network for enhanced skin lesion segmentation," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 40, no. 2, Feb. 2026, Art. no. 2550033.
[36] S. N. Srujana, N. Veena, and S. A. Karthik, "A novel approach for the detection of skin cancer using gen AI," in *Proc. IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECCT)*, Jul. 2025, pp. 1–6.
[37] M. A. Qureshi, M. Deriche, A. Beghdadi, and M. Mohandes, "An information based framework for performance evaluation of image enhancement methods," in *Proc. Int. Conf. Image Process. Theory, Tools Appl. (IPTA)*, Nov. 2015, pp. 519–523.
[38] A. Beghdadi, M. A. Qureshi, S. A. Amirshahi, A. Chetouani, and M. Pedersen, "A critical analysis on perceptual contrast and its use in visual information analysis and processing," *IEEE Access*, vol. 8, pp. 156929–156953, 2020.
[39] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, no. 1, Aug. 2018, Art. no. 180161.
[40] N. C. F. Codella, V. Rotemberg, and P. Tschandl. (2019). *ISIC 2019 Challenge: Skin Lesion Analysis Towards Melanoma Detection*. International Skin Imaging Collaboration (ISIC) Challenge. [Online]. Available: https://challenge.isic-archive.com/landing/2019/
[41] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed., Upper Saddle River, NJ, USA: Prentice-Hall, 2008.
[42] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[43] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 1–11.
[44] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.
[45] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5967–5976.



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